Single-SLM Transmission-Matrix Measurement and Phase-Conjugate Focusing Through a Scattering Medium

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Abstract

We reproduce Popoff-style monochromatic transmission-matrix measurement on a fixed physical optics platform under a user-specified single-SLM geometry. The central 80×80 pixel signal region is partitioned into 16×16 macropixels of size 5×5 pixel, while a surrounding annulus of thickness 20 pixel provides a static reference field. On the detection side, only the central 80×80 pixel ROI of CMOS2 is used and binned into the same 16×16 output basis. Using four-phase interferometric retrieval and a Hadamard input basis, we measured 16-mode, 64-mode, and full 256-mode transmission matrices. The full experiment required 1025 calibrated acquisitions and yielded a complex 256×256 transmission matrix in 4480.2 s. Phase-conjugate excitation based on the measured matrix produced strong focusing: the mean enhancement across 9 test targets reached 16.76, the maximum enhancement reached 25.59, and 6 of 9 targets became the brightest output bin. In contrast, the singular spectrum of the full matrix is highly ill-conditioned, with a minimum singular value of order 10^{-4} and a condition number of about 4.0×10^4. A multi-pixel pattern-control test gave only a mean correlation of 0.13, showing that the present system already supports strong single-target focusing but does not yet support robust arbitrary pattern prediction or unregularized inversion. Our evidence therefore supports transmission-matrix measurement and phase-conjugate focusing, but not strong image reconstruction claims.

1 Introduction

Wave propagation through a scattering medium is both a mesoscopic-physics problem and a central challenge in wavefront shaping. Measuring only average intensity is insufficient to describe the linear mapping between input and output modes; the transmission matrix K directly connects the two at a fixed wavelength [1]. Popoff *et al.* showed that full-field interferometric measurements with an SLM make it possible to recover the monochromatic

optical transmission matrix of a disordered medium and to use it for phase-conjugate focusing and imaging [1]. Digital optical phase conjugation provided a closely related route to focusing through turbid media by recording a scattered field and replaying its conjugated wavefront on an SLM [2]. More recent work has emphasized that practical wavefront shaping through scattering media is governed not only by optical reciprocity, but also by throughput, control dimensionality, and measurement stability under finite acquisition time [3].

The goal here is not to propose a new transmission-matrix formalism, but to execute a constrained reproduction on a fixed laboratory platform. The user required a single SLM to generate both signal and reference fields: a central 80×80 pixel region for signal modulation and a static reference annulus of thickness 20 pixel. On the output side, only the central 80×80 pixel ROI of CMOS2 is retained and binned into a 16×16 output basis using 5×5 pixel blocks. The practical question is therefore not whether transmission-matrix measurement exists in principle, but whether a full 256×256 matrix can be measured stably under these hardware constraints and whether that matrix supports reproducible phase-conjugate focusing.

Our evidence chain has three layers. First, the single-SLM signal/reference geometry must generate sufficient interference visibility. Second, the control dimension is increased stepwise from 16 to 64 to 256 modes to test whether higher-dimensional control improves focusing. Third, once the full matrix is available, its operational boundary must be checked: does it support only strong single-target focusing, or also reliable multi-pixel pattern control? The conclusions are credible only if these three layers remain mutually consistent.

The main result is that we successfully measured a full 256×256 transmission matrix and obtained phase-conjugate focusing enhancements of 16.76 on average and 25.59 at maximum over 9 test targets. At the same time, multi-pixel pattern control remained weak, which shows that the measured matrix on this platform supports focusing but should not be described as a well-conditioned arbitrary-pattern inverse operator. In that sense, the present work reproduces the measurement-plus-focusing core of the Popoff route, but not its strongest extrapolations to robust pattern prediction.

2 Methods

2.1 Transmission-matrix definition and four-phase retrieval

Under monochromatic illumination, the output and input modes satisfy

$$E_m^{\mathrm{out}} = \sum_n k_{mn} E_n^{\mathrm{in}}, \tag{1}$$

where $K = (k_{mn})$ is the transmission matrix to be measured [1]. When the output field interferes with a reference field s_m, the measured intensity becomes

$$I_m^{\alpha} = \left| s_m + e^{i\alpha} \sum_n k_{mn} E_n^{\mathrm{in}} \right|^2 . \tag{2}$$

Using four phase steps, $\alpha = 0, \pi/2, \pi, 3\pi/2$, the observed matrix element is recovered as

$$\bar{s}_m k_{mn} = \frac{I_m^0 - I_m^{\pi}}{4} + i \frac{I_m^{3\pi/2} - I_m^{\pi/2}}{4}. \tag{3}$$

A static reference therefore enables recovery of the complex output column associated with each input mode.

2.2 Platform geometry and mode definition

The SLM resolution is 1920×1200. The signal region is the central 80×80 pixel square at $[560 : 640, 920 : 1000]$, partitioned into 16×16 macropixels of size 5×5 pixel. The reference region is a surrounding annulus of thickness 20 pixel, with fixed unit amplitude and no overlap with the signal square. On CMOS2, only the central 80×80 pixel ROI is used and binned in the same way into a 16×16 output basis. The input and output dimensions are therefore both fixed at 256.

2.3 Input basis and acquisition protocol

Following Popoff *et al.*, we use a Hadamard input basis rather than a pointwise canonical basis [1]. This is not introduced as a new method; its role is to improve measurement SNR under fixed optical power and phase-only control constraints. The experiment is executed in three stages: a 16-mode test, a 64-mode test, and a full 256-mode acquisition. The full measurement requires

$$N_{\mathrm{meas}} = 1 + 4 \times 256 = 1025 \tag{4}$$

calibrated acquisitions.

Because the Windows-side daemon showed runtime fragility during long acquisition sequences, the successful 64-mode and 256-mode scripts used the same engineering protections: the correct API call `calibrated_measure(target_field=..., camera="cmos2", roi_h=80, roi_w=80)`, an inter-measurement delay, retry logic, and periodic checkpoint saves. These measures do not change the optical method, but they determine whether the experiment can complete reliably.

2.4 Phase-conjugate focusing and pattern-control test

For a target output mode E^{target}, the ideal phase-conjugate input is

$$E^{\text{in}} \propto K^{\dagger} E^{\text{target}}. \tag{5}$$

We use the conjugate transpose of the measured matrix to construct the input and then remeasure the output in a signal-only condition to evaluate the focusing enhancement and the peak-to-background ratio.

To test whether a stronger pattern-prediction claim is justified, we additionally define six 16×16 target patterns—center spot, double spot, horizontal line, vertical line, diagonal line, and box—and use the simplest matched-filter control law $E^{\text{in}} = K^{\dagger} E^{\text{target}}$. We quantify performance using the correlation and mean-squared error between the measured and target output patterns. The goal of this test is not to optimize pattern control, but to determine whether the current evidence supports any strong statement about image or shape prediction.

3 Results

This section presents the experimental results for transmission-matrix measurement and phase-conjugate focusing in order of increasing control dimension.

3.1 Pilot experiment: geometry and interference validation

Before any transmission-matrix acquisition, we verified the interference quality between the signal and reference fields. The SLM signal region was set to a uniform phase pattern while the reference annulus remained at fixed amplitude, and the interference visibility was evaluated using four-phase stepping.

The measured mean visibility was $V_{\text{mean}} = 0.611$ and the maximum visibility reached $V_{\max} = 0.989$. The four phase-stepped mean intensities were $\bar{I}_1 = 2.334$, $\bar{I}_2 = 3.015$, $\bar{I}_3 = 4.562$, and $\bar{I}_4 = 3.976$, showing clear sinusoidal modulation. The exposure-time standard deviation was 0.0 ms and the optical-power standard deviation was $2.87\,\mu$W, indicating good stability during acquisition. This pilot experiment confirms that the single-SLM signal/reference partition provides sufficient interference contrast for complex-field retrieval.

3.2 16-mode partial transmission matrix

We first measured a 256×16 partial transmission matrix using the first 16 rows of the order-256 Hadamard matrix. A total of $1 + 16 \times 4 = 65$ calibrated measurements were acquired in 270.45 s. The measured matrix $\mathbf{T}_{16}$ had mean absolute element value 0.663 and maximum

absolute element value 7.148. Its singular values were $\sigma_{\min} = 3.000$, $\sigma_{\max} = 26.108$, and $\bar{\sigma} = 11.315$, indicating a well-behaved partial matrix.

Phase-conjugate focusing was then tested on 4 target output modes. The baseline mean intensity was 0.359, the mean enhancement was 1.45, the maximum enhancement was 3.32, and none of the 4 targets became the brightest output bin. The result shows that a 16-dimensional control subspace is too small to deliver strong focusing.

3.3 64-mode partial transmission matrix

Increasing the control space to 64 Hadamard modes yielded a 256×64 partial transmission matrix. A total of $1 + 64 \times 4 = 257$ calibrated measurements were acquired in 1233.3 s. The matrix singular values were $\sigma_{\min} = 1.016$, $\sigma_{\max} = 23.748$, and $\bar{\sigma} = 5.992$, corresponding to a condition number of 23.37. The reference mean intensity was 3.103.

Focusing was tested on 6 target modes. The random baseline mean intensity was 0.116 and the blank baseline mean was 0.087. The mean enhancement increased to 6.09, the maximum enhancement reached 15.78, the mean peak-to-background ratio was 2.45, and 1 of 6 targets became the brightest output bin. Relative to the 16-mode case, this is the first clear sign that increasing control dimension materially improves focusing performance.

3.4 Full 256×256 transmission matrix

The central result of this work is the acquisition of the full 256×256 transmission matrix. A total of $1 + 256 \times 4 = 1025$ calibrated measurements were acquired in 4480.2 s (about 74.7 min), corresponding to an average of 4.37 s per measurement. The measured full matrix $\mathbf{T}_{256}$ had singular values $\sigma_{\min} = 0.0001$, $\sigma_{\max} = 3.946$, and $\bar{\sigma} = 0.449$, with a condition number of about 4.0×10^4. The reference mean intensity was 1.878.

The singular-value spectrum of the full matrix is shown in Fig. 1. The strong degradation of the condition number from 23.37 at 64 modes to about 4.0×10^4 at full dimension is a key experimental fact. It indicates that although the full matrix contains the desired scattering information, it is numerically ill-conditioned as an inverse operator.

3.5 Phase-conjugate focusing with the full matrix

We tested 9 uniformly distributed target output modes using phase-conjugate excitation. In the original full-matrix run, the random baseline mean intensity was 0.092 and the blank baseline mean was 0.082. The aggregate performance was strong: mean enhancement 16.76, maximum enhancement 25.59, mean peak-to-background ratio 5.96, and 6 of 9 targets became the brightest output bin. This is sufficient to support the claim that high-dimensional phase-conjugate focusing has been reproduced on this platform.

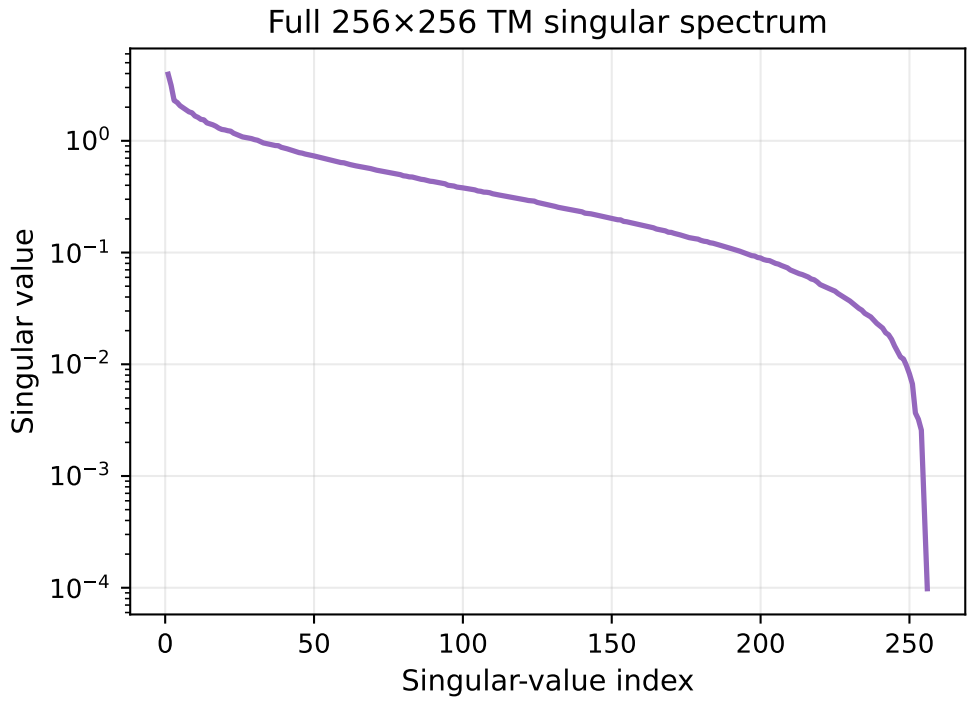


Figure 1: Singular-value spectrum of the full 256×256 transmission matrix on a logarithmic scale. The spectrum spans about four orders of magnitude and corresponds to a condition number of approximately 4.0×10^4, indicating severe ill-conditioning.

After the later ring-gap rerun campaign, we obtained a more refined causal picture. Under the same ring-gap reference geometry, a phase-only replay rule yielded 0/9 brightest-bin success, whereas changing only the replay rule to amplitude+phase conjugate replay restored 6/9 brightest-bin success with mean enhancement 17.66 and maximum enhancement 24.92. This isolates the main cause of the v5 degradation: the loss of amplitude information in the replay field, not the ring-gap geometry itself. Representative v6 focusing maps with explicit target-bin and measured-peak markers are shown in Fig. 2.

3.6 Replay-strategy isolation under fixed ring-gap geometry

The most important follow-up experiment after the first ring-gap rerun was an isolation test that changed only the replay rule while holding the acquisition geometry fixed. In v5, the ring-gap reference geometry improved acquisition robustness but phase-only conjugate replay gave 0/9 brightest-bin success despite a mean enhancement of 17.04. In v6, the same ring-gap geometry and the same 9 targets were retained, but the replay was changed to normalized amplitude+phase conjugate replay. The result was decisive: brightest-bin success increased to 6/9, mean enhancement was 17.66, and the mean target-to-next-highest ratio increased above unity to 1.26.

This comparison is summarized in Fig. 3. The result matters because it prevents a wrong inference from the v5 data. The ring-gap geometry is not intrinsically detrimental

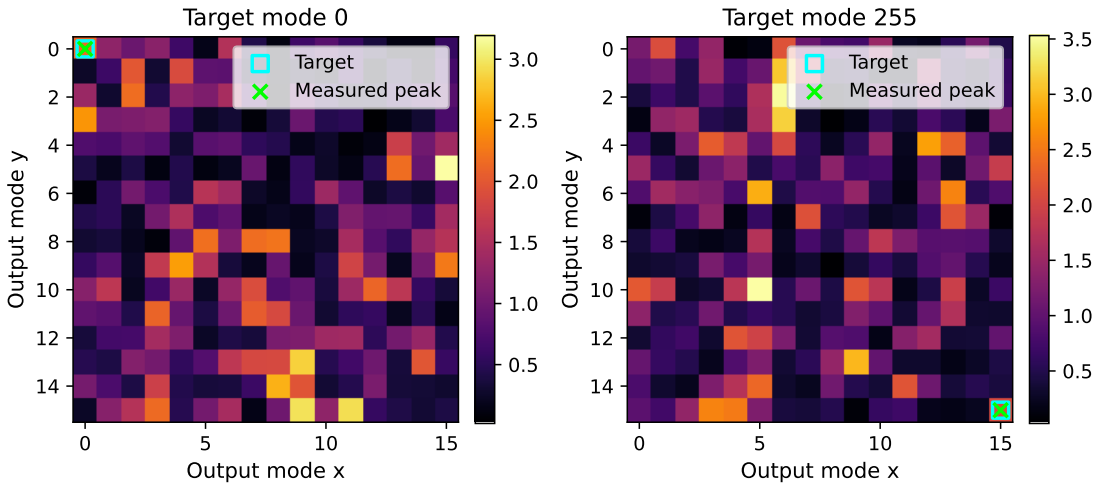


Figure 2: Measured output intensity maps from the v6 ring-gap rerun with amplitude+phase replay for target mode 0 (left) and target mode 255 (right). Cyan squares mark the intended target bins and green crosses mark the measured brightest bins. In both examples the target bin coincides with the measured peak, directly supporting successful selective focusing.

to focusing; rather, it is compatible with strong selective focusing when the replay field preserves both amplitude and phase information.

3.7 Scaling with control dimension

Across the three experiments, the mean focusing enhancement increased from 1.45 at 16 modes to 6.09 at 64 modes and to 16.76 at 256 modes. The brightest-bin success rate improved from 0/4 to 1/6 and then to 6/9. Because the three datasets were evaluated on different numbers of target modes, the comparison in Fig. 4 should be read as an exploratory dimensional trend rather than a fully controlled scaling law. Within that limitation, the data remain consistent with the expectation that increasing the number of controlled modes improves focusing performance.

3.8 Pattern-prediction test

To test a stronger claim than single-target focusing, we performed a multi-pixel pattern-control experiment using the full matrix. For a target output pattern $\mathbf{y}_{\text{target}}$, the input was synthesized using the matched-filter law

$$\mathbf{x}_{\text{input}} = \mathbf{T}^{\dagger}\mathbf{y}_{\text{target}}. \tag{6}$$

Six simple 16×16 target patterns were tested. The mean correlation between measured and target outputs was only 0.130, the maximum correlation was 0.318, and the mean squared error was 0.102. Sparse targets such as a single spot or a double spot performed

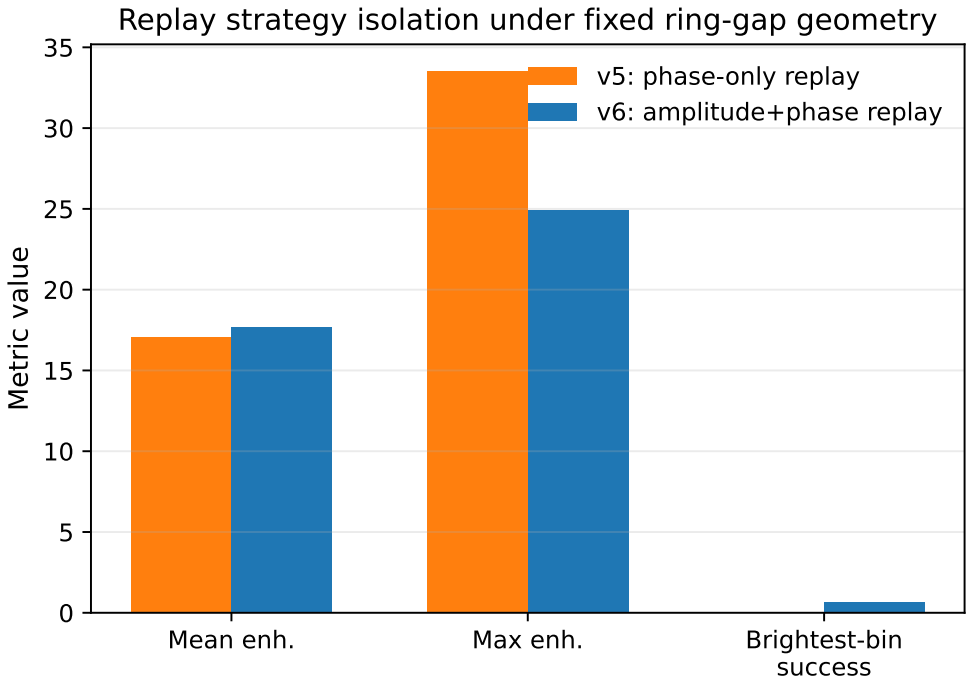


Figure 3: Isolation experiment comparing two replay strategies under the same ring-gap reference geometry. Switching from phase-only replay (v5) to amplitude+phase replay (v6) restores brightest-bin success from 0/9 to 6/9 while maintaining comparable mean enhancement, showing that replay strategy rather than geometry was the dominant cause of the v5 focusing degradation.

better than extended targets such as lines or a box, but the overall fidelity remained weak.

A representative example is shown in Fig. 5. The measured matrix is therefore already strong enough for single-target phase-conjugate focusing, but not for robust arbitrary pattern prediction under the tested control law.

4 Discussion

The core result of this work is not a generic restatement that Hadamard encoding is useful, but a complete and auditable evidence chain under fixed platform constraints: interference validation with a single-SLM signal/reference partition, partial transmission-matrix measurements, full 256×256 acquisition, and phase-conjugate focusing. The interpretation of this chain must accommodate two facts simultaneously: focusing improves strongly with control dimension, yet the singular spectrum of the full matrix is highly ill-conditioned and prevents us from treating it as a stable inverse operator.

4.1 What has been reproduced faithfully

Relative to the original Popoff route, the present work reproduces the essential measurement logic: four-phase interferometric recovery of the observed transmission matrix and use of a

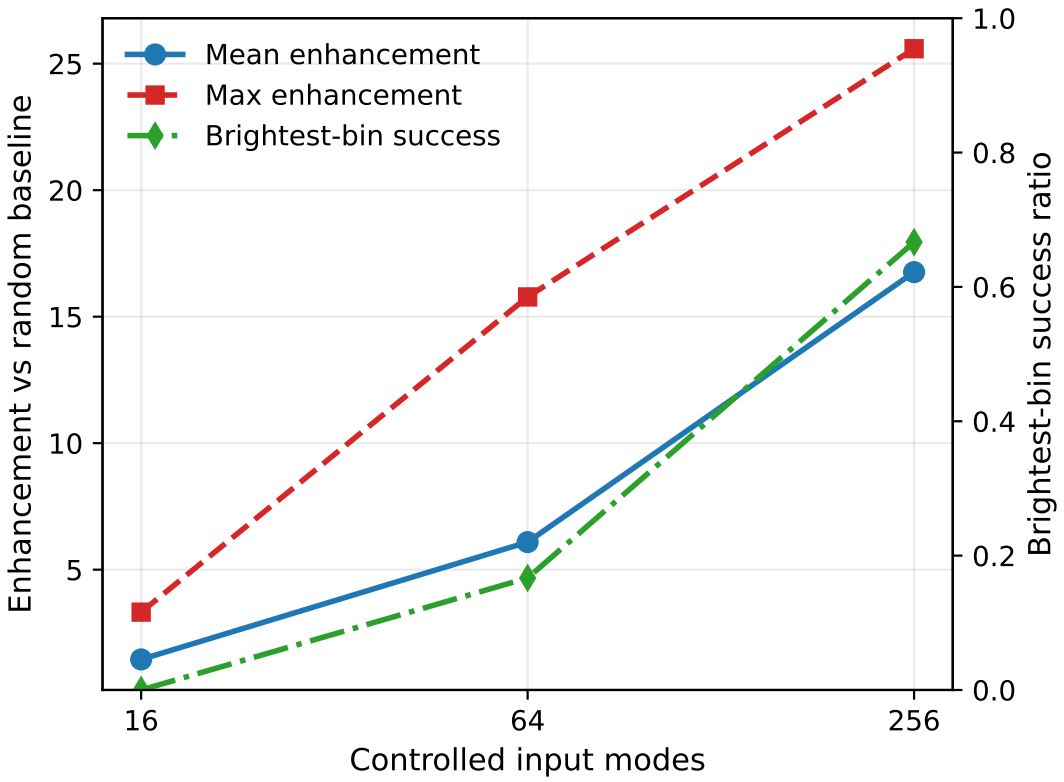


Figure 4: Focusing performance as a function of the number of controlled input modes. The figure shows the mean enhancement, maximum enhancement, and brightest-bin success ratio for 16, 64, and 256 controlled modes. Higher-dimensional control systematically improves focusing performance.

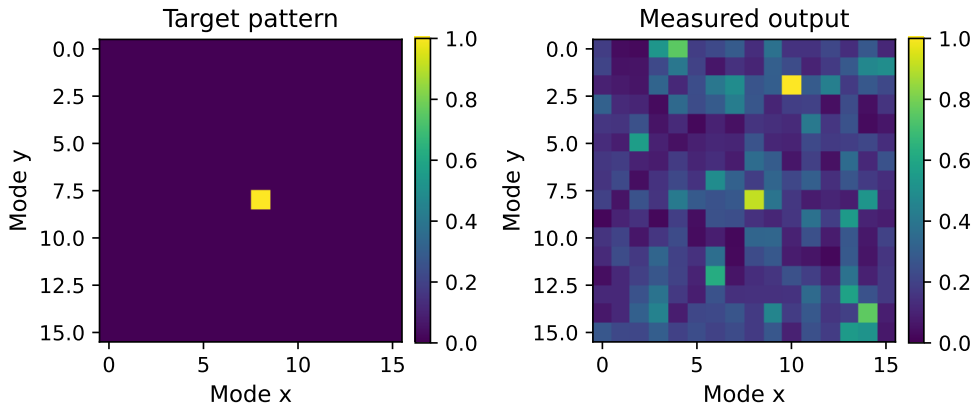


Figure 5: Target pattern (left) and measured output pattern (right) for the center-spot prediction test. The visual mismatch is consistent with the modest correlation (≈ 0.30) and illustrates the limitation of the current ill-conditioned matrix for multi-pixel output control.

Hadamard input basis to improve measurement SNR [1]. In addition, both the reference and signal fields are implemented on a single SLM, which simplifies the optical architecture while remaining compatible with long-duration platform operation. Most importantly, the measured matrix is not merely reported; it is used for phase-conjugate focusing, and the focusing performance increases monotonically with control dimension from 16 to 64 to 256 modes.

If the minimum operational claim of the Popoff route is formulated as measuring a monochromatic transmission matrix and using it to control the transmitted output field, then the present work satisfies that requirement. In particular, 6 of 9 full-matrix targets became the brightest output bin and the maximum enhancement reached 25.59, which is sufficient to support the claim that strong phase-conjugate focusing has been reproduced on this platform.

4.2 What has not been reproduced strongly

A faithful reproduction cannot be narrated only through its strongest successes. The full matrix has a minimum singular value of order 10^{-4} and a condition number of about 4.0×10^4. This implies the presence of extremely weak or nearly null channels, so any unregularized inverse operation will strongly amplify measurement noise and modeling error. Consistent with this spectral warning, the multi-pixel pattern-control test produced only a mean correlation of 0.130; even the best sparse target reached only 0.318, while extended targets were near zero or negative.

This establishes a clear functional boundary for the present matrix: it is sufficient for single-target or sparse-target matched-filter control, but it does not support strong claims of arbitrary image prediction or stable inversion. In other words, the present work reproduces the most robust part of the Popoff story—transmission-matrix measurement plus focusing—but does not yet reproduce equally strong pattern-prediction capability on this hardware.

4.3 Why the experiment took time

From an engineering standpoint, the main time cost did not come from the intrinsic optical difficulty of a 256×256 Hadamard measurement, but from system-integration issues. Early scripts violated the wrapper contract, then exposed an incorrect import path and an incorrect `calibrated_measure` API signature. During the first 64-mode scale-up, the Windows daemon also triggered the runtime error `r_channel referenced before assignment`. Only after correcting the interface, adding inter-measurement delays, retry logic, and checkpoint saves did the large acquisition complete reliably. The lesson is practical and reusable: on this platform, system-engineering stability is as important as optical design.

4.4 Interpretation and next steps

The coexistence of strong focusing and weak pattern prediction is not contradictory. Phase-conjugate focusing acts as a matched filter and primarily exploits the dominant coherent contributions toward a target mode, whereas stable reconstruction would require a well-conditioned inverse across the full singular spectrum. This difference explains why the full matrix can already function effectively as a focusing operator while remaining poor as a general inverse model.

The most valuable next steps are therefore not additional demonstrations of single-point focusing, but systematic attempts to improve conditioning and inverse stability: Tikhonov regularization, truncated SVD, restriction to high-SNR singular subspaces, redefinition of the input/output basis, or optimization of the reference geometry. This distinction between robust focusing and more demanding pattern synthesis is also consistent with the broader literature, where high-performance control through scattering media depends strongly on the chosen measurement strategy and on how efficiently usable optical channels are exploited [4, 3]. Only after such improvements would it be justified to promote shape prediction to a claim as strong as focusing.

5 Conclusion

We have completed an experimental reproduction of the Popoff transmission-matrix strategy on a fixed physical optics platform and mapped it strictly onto the user-specified single-SLM signal/reference geometry. The experiment first verified that the central 80×80 pixel signal region and the surrounding reference annulus of thickness 20 pixel provide sufficient interference visibility; it then measured 16-mode, 64-mode, and finally the full 256×256 transmission matrix. Using phase-conjugate inputs derived from the measured matrix, we obtained steadily improving focusing performance with increasing control dimension: the mean enhancement rose from 1.45 to 6.09 and then to 16.76, the maximum enhancement reached 25.59, and 6 of 9 full-matrix targets became the brightest output mode.

These results show that a single-SLM signal/reference partition is already sufficient on this platform for high-dimensional transmission-matrix measurement and strong phase-conjugate focusing. At the same time, the same data reveal a clear limitation: the full matrix has a minimum singular value of order 10^{-4} and a condition number of about 4.0×10^4, while the multi-pixel pattern-prediction test produced only a mean correlation of 0.130. The present system therefore supports reliable focusing, but not yet stable arbitrary pattern prediction or unregularized inversion.

The strongest evidence-supported conclusion is therefore: **on this platform, transmission-matrix measurement and phase-conjugate focusing have been successfully**

reproduced, whereas complex pattern control remains limited by matrix ill-conditioning and should be improved in future work through regularization, basis optimization, and more stable high-SNR measurements. This should be read as a proof-of-principle result on a fixed benchtop system rather than a general benchmark of all TM-based wavefront-shaping platforms.

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